

PilgrimIQ

1. Project Overview

PilgrimIQ: Personalized Pilgrimage Decision Support System is a web-based platform designed to enhance the safety, health, and overall pilgrimage experience through intelligent decision support and personalized travel assistance. The system assists individual pilgrims and family groups by providing AI-based health assessment, medical report analysis, journey planning, accommodation recommendations, transportation guidance, food recommendations, and personalized travel support before, during, and after the pilgrimage. It utilizes Artificial Intelligence (AI), Machine Learning (ML), Optical Character Recognition (OCR), and Natural Language Processing (NLP) to analyze user health information, uploaded medical reports, weather conditions, crowd prediction, waiting time estimation, terrain difficulty, and journey details. Based on this analysis, the system evaluates travel readiness, generates a Pilgrim Safety Index (PSI), and provides personalized recommendations such as diet plans, hydration guidance, medicine reminders, accommodation suggestions, food recommendations, transportation options, safe route recommendations, and travel alternatives. During the pilgrimage, the platform provides intelligent journey assistance through weather alerts, health safety notifications, AI chatbot support, emergency services, nearby healthcare facilities, accommodation details, food facilities, and transportation information, creating a secure and intelligent decision support ecosystem for pilgrims.

2. System Scope

The proposed system is developed as an academic project that can later be expanded into a full-scale implementation.

Current Scope

- User registration and authentication
- Individual and family profile management
- Health assessment
- AI-based medical report analysis (OCR & NLP)
- Travel readiness assessment
- Pilgrim Safety Index (PSI) generation
- Journey planning
- Pilgrimage center information
- Weather information integration
- Accommodation recommendations
- Food recommendations
- Transportation information

- Personalized recommendations
- Nearby hospitals and emergency services
- Administration management

Future Enhancements

- AI Chat Assistant
- Intelligent Journey Assistance
- Crowd Prediction
- Waiting Time Prediction
- Personalized Journey Timeline
- Dynamic Journey Risk Reassessment
- Explainable AI (XAI)
- Mobile Application
- Multilingual Support
- Adaptive Machine Learning
- Online Accommodation Booking
- Transportation Ticket Integration

3. Target Audience

The primary users of the system include:

- Individual Pilgrims
- Family Pilgrims
- General Physicians
- System Administrators

4. User Roles

• Administrator

Manages users, physicians, pilgrimage center information, accommodation and food information, AI datasets, system analytics, reports, and overall platform operations.

• Pilgrim (Individual / Family)

Creates and manages health and journey profiles, uploads medical reports, receives AI-based travel readiness assessment, Pilgrim Safety Index (PSI), personalized recommendations, accommodation suggestions, food recommendations, transportation guidance, and journey assistance.

- **General Physician**

Reviews AI-generated health assessments for high-risk pilgrims, verifies medical reports, approves or postpones travel, provides medical recommendations, and refers pilgrims to specialists when required.

5. Modules

1. **User Authentication & Profile Management Module** – Registration, login, secure authentication, and individual/family profile management.
2. **Health Assessment Module** – Health profile creation, medical history, BMI calculation, disease, allergy, medication, and emergency contact management.
3. **Medical Report Analysis Module** – OCR-based medical report extraction, NLP analysis, AI health assessment, and travel readiness evaluation.
4. **Physician Review Module** – AI report verification, travel approval, medical recommendations, postponement, and specialist referral.
5. **Pilgrimage Center Information Module** – Pilgrimage center details, maps, operating hours, climate, terrain, dress code, visitor guidelines, nearby accommodation, food facilities, and transportation information.
6. **Journey Planning Module** – Destination selection, travel schedule, transport details, travel duration, walking distance, climbing information, family journey planning, and estimated travel cost.
7. **AI Prediction & Decision Support Module** – Weather analysis, crowd prediction, waiting time estimation, terrain analysis, health risk prediction, travel readiness assessment, and Pilgrim Safety Index (PSI) generation.
8. **Personalized Recommendation Module** – Diet recommendations, hydration guidance, medicine reminders, AI packing checklist, accommodation suggestions, food recommendations, transportation recommendations, safe route recommendations, and travel alternatives.
9. **Journey Assistance Module** – Live weather alerts, journey progress tracking, health safety alerts, AI chatbot, emergency assistance, nearby services, accommodation guidance, transportation updates, and journey risk reassessment.
10. **Nearby Services & Emergency Module** – Hospitals, pharmacies, medical camps, restaurants, accommodation facilities, toilets, drinking water, parking facilities, transport hubs, emergency contacts, and Google Maps navigation.
11. **Feedback & Adaptive Learning Module** – Journey feedback, satisfaction rating, health status, recommendation feedback, and AI model improvement.
12. **Administration Module** – User management, physician management, pilgrimage center management, accommodation and food information management, AI dataset management, AI model monitoring, system analytics, and reports.

6. System Ownership

The proposed system is developed as an academic project under the **Department of Computer Applications**.

System Owner:

Academic Institution (Amal Jyothi College of Engineering, Kanjirappally)

Developed by:

Arsha Karma

Final Year Integrated MCA Student

Future ownership may be transferred to pilgrimage authorities, tourism departments, or religious organizations for practical implementation.

7. Questionnaire for Data Collection

Target Users: Pilgrims

1. Have you previously undertaken a pilgrimage?

Yes / No

2. Do you usually travel individually or with your family?

Individual / Family

3. Do you have any existing medical conditions?

Yes / No

4. Would you upload your medical reports if it helps assess your travel readiness?

Yes.

5. Which information do you consider most important before starting a pilgrimage?

Weather, Crowd Level, Waiting Time, Health Risk, Route Information, Accommodation, Food Facilities, Transportation.

6. Would personalized recommendations (diet, hydration, medicines, accommodation, food, transportation, safe routes) help during your pilgrimage?

Yes.

7. Which journey assistance feature would you find most useful?

Weather Alerts, Emergency Services, Nearby Hospitals, AI Chat Assistant, Health Reminders, Accommodation Information, Food Recommendations, Transportation Guidance.

8. Would you like to know your Pilgrim Safety Index (PSI) before beginning your journey?

Yes.

9. Do you think AI-based travel readiness assessment would improve your pilgrimage safety?

Yes.

10. Would you recommend PilgrimIQ to other pilgrims?

Yes.

8.Data Collection Contact

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